MPG No. 31.1.115 $\bar{4}3m$ [Type I, cubic] [T tensor]

* Rank 0 tensor.

$$[T]$$

$$T = \mathbb{T}_0^{(1)}$$

* Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} T_{xx} & 0 & 0 \\ 0 & T_{xx} & 0 \\ 0 & 0 & T_{xx} \end{bmatrix}$$

$$T_{xx} = \mathbb{T}_0^{(1)}$$

* Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ T_{yzx} & 0 & 0 \\ 0 & T_{yzx} & 0 \\ 0 & 0 & T_{yzx} \end{bmatrix}$$

$$T_{yzx} = \mathbb{T}_{xyz}^{(1)}$$

* Rank 3 tensor (sa). * Rank 3 tensor (a). * Rank 4 tensor (sss).

$$\begin{bmatrix} T_{xxxx} & T_{xxyy} & T_{xxyy} & 0 & 0 & 0 \\ T_{xxyy} & T_{xxxx} & T_{xxyy} & 0 & 0 & 0 \\ T_{xxyy} & T_{xxyy} & T_{xxxx} & 0 & 0 & 0 \\ 0 & 0 & 0 & T_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & T_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{yzyz} \end{bmatrix}$$

$$T_{xxxx} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} + 2\mathbb{T}_4^{(1)}$$
$$T_{xxyy} = \mathbb{T}_0^{(1)} - \mathbb{T}_4^{(1)}$$
$$T_{yzyz} = \mathbb{T}_0^{(2)} - \mathbb{T}_4^{(1)}$$

* Rank 4 tensor (ssa). * Rank 4 tensor (aas).

$$\begin{bmatrix} T_{yzyz} & 0 & 0 \\ 0 & T_{yzyz} & 0 \\ 0 & 0 & T_{yzyz} \end{bmatrix}$$

$$T_{yzyz} = \mathbb{T}_0^{(3)}$$

* Rank 4 tensor (aaa). * Rank 4 tensor (sa). * Rank 4 tensor (as). * Rank 4 tensor (s).

$$\begin{bmatrix} T_{xxxx} & T_{xxyy} & T_{xxyy} & 0 & 0 & 0 & 0 & 0 & 0 \\ T_{xxyy} & T_{xxxx} & T_{xxyy} & 0 & 0 & 0 & 0 & 0 & 0 \\ T_{xxyy} & T_{xxyy} & T_{xxxx} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & T_{yzyz} & 0 & 0 & T_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & T_{yzyz} & 0 & 0 & T_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{yzyz} & 0 & 0 & T_{yzyz} \end{bmatrix}$$

$$T_{xxxx} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} + 2\mathbb{T}_4^{(1)}$$
$$T_{xxyy} = \mathbb{T}_0^{(1)} - \mathbb{T}_4^{(1)}$$
$$T_{yzyz} = \mathbb{T}_0^{(2)} - \mathbb{T}_4^{(1)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & T_{yzyz} & 0 & 0 & -T_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & T_{yzyz} & 0 & 0 & -T_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{yzyz} & 0 & 0 & -T_{yzyz} \end{bmatrix}$$

$$T_{yzyz} = \mathbb{T}_0^{(3)}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} T_{xxxx} & 0 & 0 \\ 0 & T_{xxxx} & 0 \\ 0 & 0 & T_{xxxx} \\ 0 & 0 & T_{yyzz} \\ T_{yyzz} & 0 & 0 \\ 0 & T_{yyzz} & 0 \\ 0 & T_{yyzz} & 0 \\ 0 & 0 & T_{yyzz} \\ T_{yyzz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxxx} = 3\mathbb{T}_0^{(1)} + 2\mathbb{T}_4^{(1)}$$
$$T_{yyzz} = \mathbb{T}_0^{(1)} - \mathbb{T}_4^{(1)}$$